System Al–Cu

Solution Phases:

(stable)	Liquid, bcc-$A2$, fcc-$A1$
(metastable)	hcp-$A3$

Compounds:

(stable)	AlCu-δ, AlCu-ϵ, AlCu-η, AlCu-θ, AlCu-$\gamma(D8_3)$
	AlCu-γ(H), AlCu-ζ
(metastable)	Laves-$C14$, Laves-$C15$, Laves-$C36$

Modelling:

Liquid	:	Substitutional, Redlich-Kister
bcc-$A2$	:	Substitutional, Redlich-Kister
fcc-$A1$	:	Substitutional, Redlich-Kister
hcp-$A3$	:	Substitutional, Redlich-Kister
AlCu-δ	:	Sublattice model, $(Al,Cu)_2(Cu)_3$
AlCu-ϵ	:	Sublattice model, (Al,Cu)(Cu)
AlCu-η	:	Sublattice model, (Al,Cu)(Cu)
AlCu-θ	:	Sublattice model, $(Al)(Al,Cu)_2$
AlCu-$\gamma(D8_3)$	:	Sublattice model, $(Al)_4(Al,Cu)(Cu)_8$
AlCu-γ(H)	:	Sublattice model, $(Al)_4(Al,Cu)(Cu)_8$
AlCu-ζ	:	Stoichiometric, $(Al)_9(Cu)_{11}$
Laves-$C14$	:	Sublattice model, $(Al,Cu)_2(Al,Cu)$
Laves-$C15$	:	Sublattice model, $(Al,Cu)_2(Al,Cu)$
Laves-$C36$	:	Sublattice model, $(Al,Cu)_2(Al,Cu)$

Assessor and Date:

N. Saunders, 1991
Data relative to the Laves phases evaluated in Round II.

Thermodynamic properties of the solution and compound phases (J.mol^{-1})

Phase AlCu-δ

$$G^{\circ}(T) - 2.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 3.0\ H_{Cu}^{\circ,fcc-A1}(298.15\ K) = - 106700.0 + 3.0\ T + 2.0\ GHSER_{Al} + 3.0\ GHSER_{Cu}$$

Phase AlCu-ϵ

$$G^{\circ}(T) - H_{Al}^{\circ,fcc-A1}(298.15\ K) - H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Al{:}Cu) = - 36976 + 1.2\ T + GHSER_{Al} + GHSER_{Cu}$$

$$G^{\circ}(T) - 2.0\ H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Cu{:}Cu) = 2.0\ GBCC_{Cu}$$

$$L_{Al,Cu:Cu}^{0,AlCu-\epsilon} = 7600 - 24\ T$$

$$L_{Al,Cu:Cu}^{1,AlCu-\epsilon} = - 72000$$

Phase AlCu-η

$$G^{\circ}(T) - H_{Al}^{\circ,fcc-A1}(298.15\ K) - H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Al{:}Cu) = - 40560.0 + 3.14\ T + GHSER_{Al} + GHSER_{Cu}$$

$$G^{\circ}(T) - 2.0\ H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Cu{:}Cu) = 2.0\ GBCC_{Cu}$$

$$L_{Al,Cu:Cu}^{0,AlCu-\eta} = - 25740 - 20\ T$$

Phase AlCu-θ

$$G^{\circ}(T) - 3.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) = G(Al{:}Al) = 3.0\ GBCC_{Al}$$

$$G^{\circ}(T) - 2.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Al{:}Cu) = - 47406.0 + 6.75\ T + 2.0\ GHSER_{Al} + 1.0\ GHSER_{Cu}$$

$$L_{Al:Al:Cu}^{0,AlCu-\theta} = 2211$$

Phase AlCu-ζ

$$G^{\circ}(T) - 9.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 11.0\ H_{Cu}^{\circ,fcc-A1}(298.15\ K) = - 420000.0 + 18.0\ T + 9.0\ GHSER_{Al} + 11.0\ GHSER_{Cu}$$

Phase AlCu-$\gamma(D8_3)$

$$G^{\circ}(T) - 5.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 8.0\ H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Al{:}Al{:}Cu) = - 300716.0 + 390.0\ T - 52\ T \cdot \ln T + 5.0\ GHSER_{Al} + 8.0\ GHSER_{Cu}$$

$$G^{\circ}(T) - 4.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 9.0\ H_{Cu}^{\circ,fcc-A1}(298.15\ K) = G(Al{:}Cu{:}Cu) = - 280501.0 + 379.6\ T - 52\ T \cdot \ln T + 4.0\ GHSER_{Al} + 9.0\ GHSER_{Cu}$$

Phase AlCu–γ(H)

$$G^{o}(T) - 5.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - 8.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Al{:}Al{:}Cu) = -219258.0 - 45.5\ T + 5.0\ GHSER_{Al} + 8.0\ GHSER_{Cu}$$

$$G^{o}(T) - 4.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - 9.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Al{:}Cu{:}Cu) = -200460.0 - 58.5\ T + 4.0\ GHSER_{Al} + 9.0\ GHSER_{Cu}$$

Phase Laves-$C14$

$$G^{o}(T) - 3.0\ H_{Al}^{o,fcc-A1}(298.15\ K) = G(Al{:}Al) = 15000.0 + 3.0\ GHSER_{Al}$$

$$G^{o}(T) - 2.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Al{:}Cu) = 15000.0 + 2.0\ GHSER_{Al} + GHSER_{Cu}$$

$$G^{o}(T) - H_{Al}^{o,fcc-A1}(298.15\ K) - 2.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Cu{:}Al) = 15000.0 + GHSER_{Al} + 2.0\ GHSER_{Cu}$$

$$G^{o}(T) - 3.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Cu{:}Cu) = 15000.0 + 3.0\ GHSER_{Cu}$$

$$L_{Al,Cu:Al}^{0,Laves-C14} = -24000 + 2.44\ T$$

$$L_{Al,Cu:Cu}^{0,Laves-C14} = -24000 + 2.44\ T$$

Phase Laves-$C15$

$$G^{o}(T) - 3.0\ H_{Al}^{o,fcc-A1}(298.15\ K) = G(Al{:}Al) = 15000.0 + 3.0\ GHSER_{Al}$$

$$G^{o}(T) - 2.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Al{:}Cu) = 15000.0 + 2.0\ GHSER_{Al} + GHSER_{Cu}$$

$$G^{o}(T) - H_{Al}^{o,fcc-A1}(298.15\ K) - 2.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Cu{:}Al) = 15000.0 + GHSER_{Al} + 2.0\ GHSER_{Cu}$$

$$G^{o}(T) - 3.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Cu{:}Cu) = 15000.0 + 3.0\ GHSER_{Cu}$$

$$L_{Al,Cu:Al}^{0,Laves-C15} = -105000 - 1.615\ T$$

$$L_{Al,Cu:Cu}^{0,Laves-C15} = -105000 - 1.615\ T$$

Phase Laves-$C36$

$$G^{o}(T) - 3.0\ H_{Al}^{o,fcc-A1}(298.15\ K) = G(Al{:}Al) = 15000.0 + 3.0\ GHSER_{Al}$$

$$G^{o}(T) - 2.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Al{:}Cu) = 15000.0 + 2.0\ GHSER_{Al} + GHSER_{Cu}$$

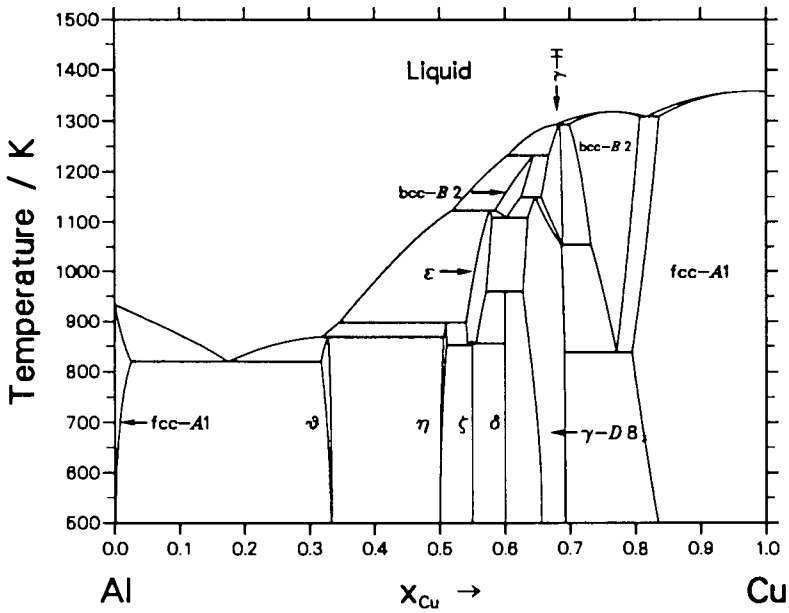
$$G^{o}(T) - H_{Al}^{o,fcc-A1}(298.15\ K) - 2.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Cu{:}Al) = 15000.0 + GHSER_{Al} + 2.0\ GHSER_{Cu}$$

$$G^{o}(T) - 3.0\ H_{Cu}^{o,fcc-A1}(298.15\ K) = G(Cu{:}Cu) =$$

$$15000.0 + 3.0 \ \mathrm{GHSER_{Cu}}$$

$$L^{0,\mathrm{Laves}-C36}_{\mathrm{Al,Cu:Al}} = -126169 + 14.61 \ T$$

$$L^{0,\mathrm{Laves}-C36}_{\mathrm{Al,Cu:Cu}} = -126169 + 14.61 \ T$$



Phase bcc–$A2$

$$L^{0,\mathrm{bcc}-A2}_{\mathrm{Al,Cu:\square}} = -73554 + 4.0 \ T$$

$$L^{1,\mathrm{bcc}-A2}_{\mathrm{Al,Cu:\square}} = 51500 - 11.84 \ T$$

Phase fcc–$A1$

$$L^{0,\mathrm{fcc}-A1}_{\mathrm{Al,Cu:\square}} = -53520 + 2 \ T$$

$$L^{1,\mathrm{fcc}-A1}_{\mathrm{Al,Cu:\square}} = 38590 - 2 \ T$$

$$L^{2,\mathrm{fcc}-A1}_{\mathrm{Al,Cu:\square}} = 1170$$

Phase liquid

$$L^{0,\mathrm{liquid}}_{\mathrm{Al,Cu}} = -66622 + 8.1\ T$$

$$L^{1,\mathrm{liquid}}_{\mathrm{Al,Cu}} = 46800 - 90.8\ T + 10\ T \cdot \ln T$$

$$L^{2,\mathrm{liquid}}_{\mathrm{Al,Cu}} = -2812$$

Table I – Invariant Reactions.

Reaction	Type	Compositions x_{Cu}			T / K
Liquid ⇌ fcc-$A1$ + AlCu-θ	Eutectic	.175	.025	.318	820.7
Liquid + AlCu-η ⇌ AlCu-θ	Peritectic	.319	.506	.329	869.0
Liquid + AlCu-ϵ ⇌ AlCu-η	Peritectic	.345	.541	.510	898.0
Liquid + bcc-$A2$ ⇌ AlCu-ϵ	Peritectic	.519	.585	.576	1124.0
Liquid + AlCu-γ(H) ⇌ bcc-$A2$	Peritectic	.604	.667	.644	1232.4
Liquid + bcc-$A2$ ⇌ AlCu-γ(H)	Peritectic	.678	.698	.684	1293.4
Liquid ⇌ bcc-$A2$ + fcc-$A1$	Eutectic	.817	.807	.837	1309.3
bcc-$A2$ ⇌ AlCu-$\gamma(D8_3)$ + fcc-$A1$	Eutectoid	.771	.691	.795	837.9
AlCu-γ(H) ⇌ AlCu-$\gamma(D8_3)$ + bcc-$A2$	Eutectoid	.688	.686	.732	1054.2
bcc-$A2$ + AlCu-γ(H) ⇌ AlCu-$\gamma(D8_3)$	Peritectoid	.625	.656	.647	1150.5
bcc-$A2$ ⇌ AlCu-ϵ + AlCu-$\gamma(D8_3)$	Eutectoid	.602	.580	.635	1109.3
AlCu-ϵ + AlCu-$\gamma(D8_3)$ ⇌ AlCu-δ	Peritectoid	.572	.628	.600	959.7
AlCu-ϵ ⇌ AlCu-ζ + AlCu-δ	Eutectoid	.555	.550	.600	855.8
AlCu-ϵ ⇌ AlCu-η + AlCu-ζ	Eutectoid	.543	.511	.550	852.4
AlCu-ζ ⇌ AlCu-ϵ	Congruent	.550	.550	...	861.0

Table II:a – Crystal Structure and Phase Description.

System	Phase	Prototype	Pearson Symbol Space Group	Sub-lattices	Comments
Al-Cu	(Al)	Cu	$cF4$ $Fm\bar{3}m$	M 4	
	(Cu)	Cu	$cF4$ $Fm\bar{3}m$	M 4	
	β	W	$cI2$ $Im\bar{3}m$	M 2	
	β_0				not considered

Table II:b – Crystal Structure and Phase Description.

System	Phase	Prototype	Pearson Symbol Space Group	Sub-lattices	Comments
	AlCu-δ		.... $R3m$		
	AlCu-η_1		$Pban$ or $Cmmm$ $oP16$ or $oC16$		high T phase
	AlCu-η_2	AlCu	$mC20$ $C2/m$		low T phase
	AlCu-θ	Al_2Cu	$tI12$ $I4/mcm$	Al 8 Cu 4	
	AlCu-δ	.....	..... $R\bar{3}m$		
	AlCu-ϵ_1	NiAs	$hP4$ $P6_3/mmc$		low T phase
	AlCu-ϵ_2	.....	.....		not considered
	AlCu-$\gamma(D8_3)$	Al_4Cu_9	$cP52$ $P\bar{4}3m$	Al_1 4 Al_2 12 Cu_1 4 Cu_2 4 Cu_3 4 Cu_4 6 Cu_5 6 Cu_6 12	
	AlCu-γ(H)	.....	....		
	AlCu-ζ_1	.....	$hP42$ $C2/m$		ζ_1
	AlCu-ζ_2	.....			ζ_2 not considered